

OS SKILL 4

Tokenizer.c:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

int main() {
    char str[100], token[20];
    int i = 0, j;

    printf("Enter command: ");
    fgets(str, sizeof(str), stdin);

    printf("\nTOKENS:\n");

    while (str[i] != '\0' && str[i] != '\n') {

        if (isspace(str[i])) {
            i++;
            continue;
        }

        if (strchr("<>\"", str[i])) {
            printf("DELIMITER : %c\n", str[i]);
            i++;
        }
        else {
            j = 0;

            while (str[i] &&
                    !isspace(str[i]) &&
                    !strchr("<>\"", str[i])) {
                token[j++] = str[i++];
            }

            token[j] = '\0';
            printf("WORD      : %s\n", token);
        }
    }

    return 0;
}
```

```
nishanth@LAPTOP-6Q3ON07H:~/skill4$ gcc tokenizer.c -o tokenizer
okenizernishanth@LAPTOP-6Q3ON07H:~/skill4$ ./tokenizer
Enter command: cat file.txt | grep hello > output.txt

TOKENS:
WORD      : cat
WORD      : file.txt
DELIMITER : |
WORD      : grep
WORD      : hello
DELIMITER : >
WORD      : output.txt
```

Parser.c:

```

int main() {
    char cmd[100];

    printf("Enter command: ");
    fgets(cmd, sizeof(cmd), stdin);

    cmd[strcspn(cmd, "\n")] = '\0';

    if (strlen(cmd) == 0) {
        printf("Error: Empty command\n");
        return 0;
    }

    int len = strlen(cmd);

    if (cmd[0] == '|' || cmd[len - 1] == '|') {
        printf("Syntax Error: Invalid pipe\n");
        return 0;
    }

    if (cmd[len - 1] == '>' || cmd[len - 1] == '<') {
        printf("Syntax Error: Filename missing\n");
        return 0;
    }

    if (strstr(cmd, "|")) {
        printf("Syntax Error: Empty command between pipes\n");
        return 0;
    }

    printf("Syntax Valid!\n");

    printf("\nPARSE / EXECUTION STRUCTURE:\n");

    char *part = strtok(cmd, "|");

    while (part != NULL) {
        printf("COMMAND -> %s\n", part);
        part = strtok(NULL, "|");

        if (part != NULL)
            printf("    |\n    PIPE\n    |\n");
    }

    return 0;
}

```

```
nishanth@LAPTOP-6Q3ON07H:~/skill4$ nano parser.c
nishanth@LAPTOP-6Q3ON07H:~/skill4$ gcc parser.c -o parser
./parser
Enter command: ls -l | grep txt
Syntax Valid!
```

```
PARSE / EXECUTION STRUCTURE:
COMMAND -> ls -l
|
PIPE
|
COMMAND -> grep txt
```